INSIGHTVISTA 2026

OLYMPIC ATHLETE EVENT ANALYTICS

From raw records to business-ready insights

GROUP • Analytics Squad

Member 1 | PEN: 240840116043

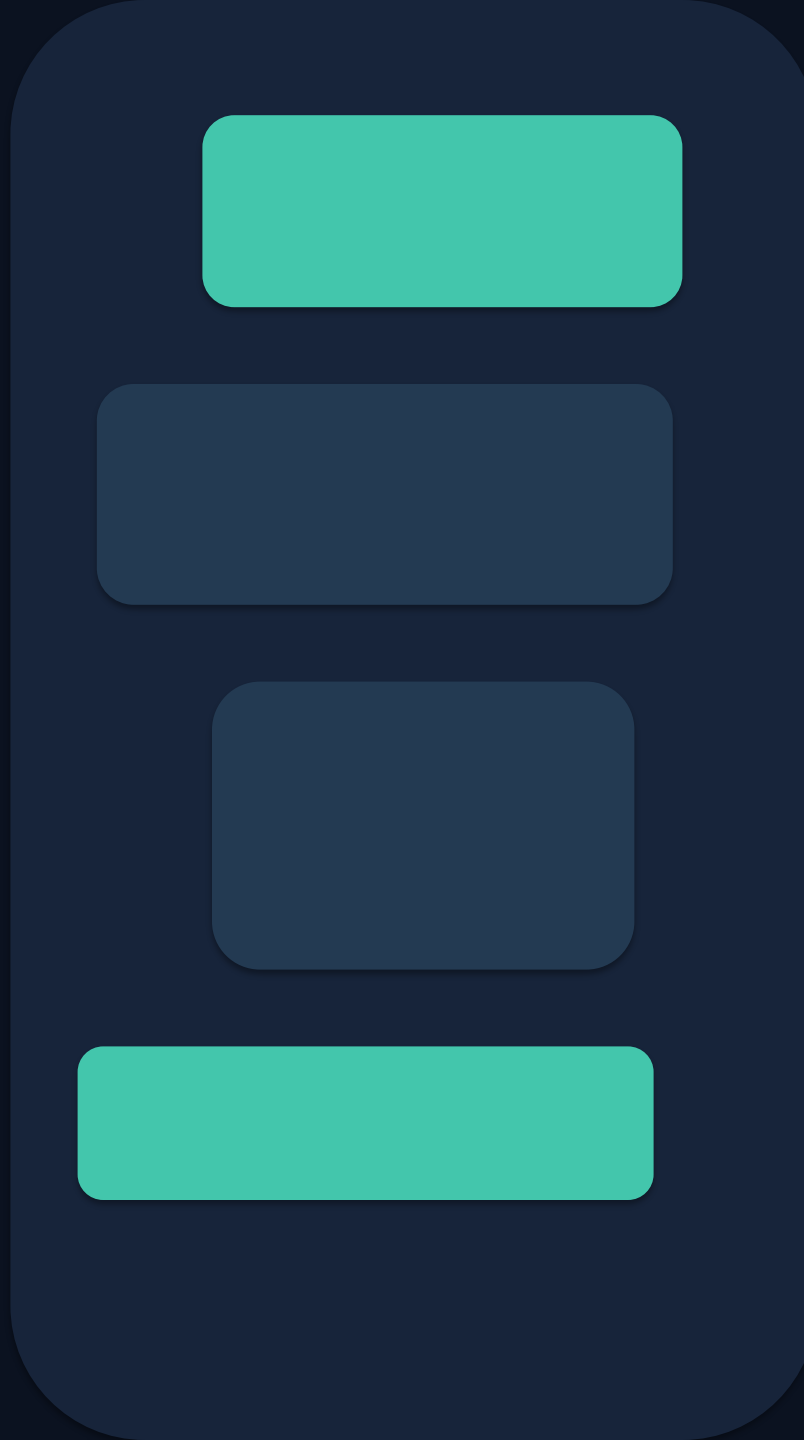
Member 2 | PEN: 240840116049

Member 3 | PEN: 240840116127

Member 4 | PEN: 250843116017

Member 5 | PEN: 240840116095

Dataset: Olympic athlete events • Coverage: 1896–2016



01 | Raw Data Observation

Understanding the dataset before analysis

ROWS	COLUMNS	SPORTS	TEAMS	YEARS
269,731	13	66	1,184	1896–2016

- 13 fields capture athlete identity, demographics, team/NOC, year, season, host city, sport, event and medal outcome.
- The dataset spans 1896–2016 and contains both Summer and Winter Olympic records.
- Most records are from Summer Olympics (221,167); Winter records: 48,564.
- Medal values include Gold, Silver, Bronze and No Medal; each row represents an athlete-event participation record.

02 | Steps Performed on the Raw Data

A reproducible data preparation workflow

1. **Load** — Imported the CSV and inspected dimensions, column names and data types.
2. **Profile** — Checked missing values, duplicate rows and category distributions.
3. **Validate** — Reviewed numeric fields (Age, Height, Weight, Year) and category fields.
4. **Standardize** — Used consistent column labels and interpreted “No Medal” as a valid outcome, not missing data.
5. **Prepare** — Grouped records by sport, year, team and medal outcome for visualization.

03 | Final Clean Dataset

Quality checks and analytical-ready structure

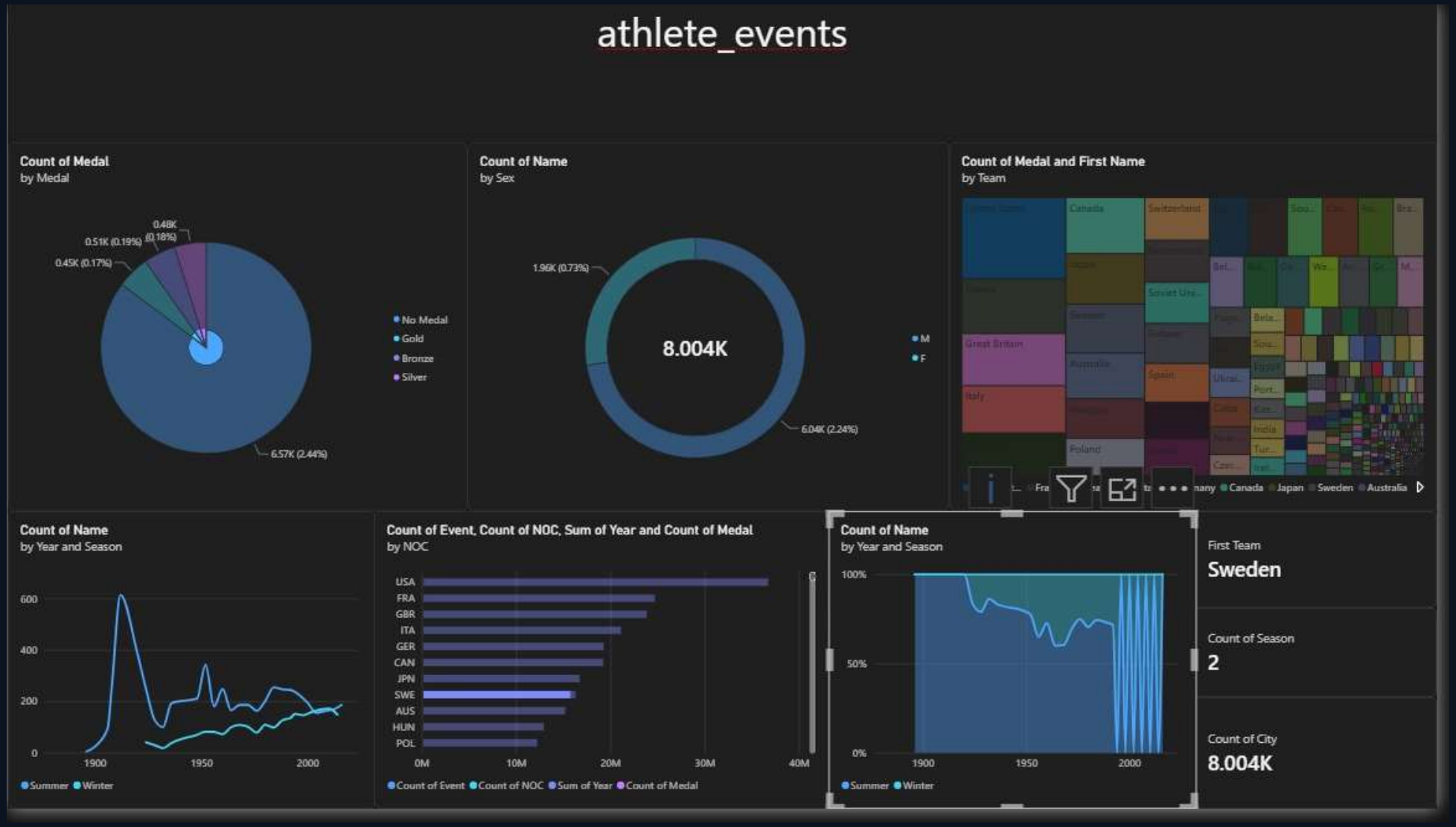
ROWS RETAINED	MISSING CELLS	DUPLICATE ROWS	MEDAL RECORDS
269,731	0	0	39,772

- The supplied file is already a clean dataset: all 269,731 rows have values in all 13 columns.
- No exact duplicate rows were detected, so no records needed to be removed for duplication.
- Numeric fields are ready for summary statistics; categorical fields are ready for grouping and comparison.
- “No Medal” was preserved as a meaningful category. Medal-winning records are counted separately when needed.
- Important interpretation: medal counts here are athlete-event records, not necessarily unique physical medals.

Output: a validated, analysis-ready table for dashboarding and recommendations.

04 | Visualization Dashboard

Actual Power BI dashboard provided by Analytics Squad



05 | Dashboard Observations

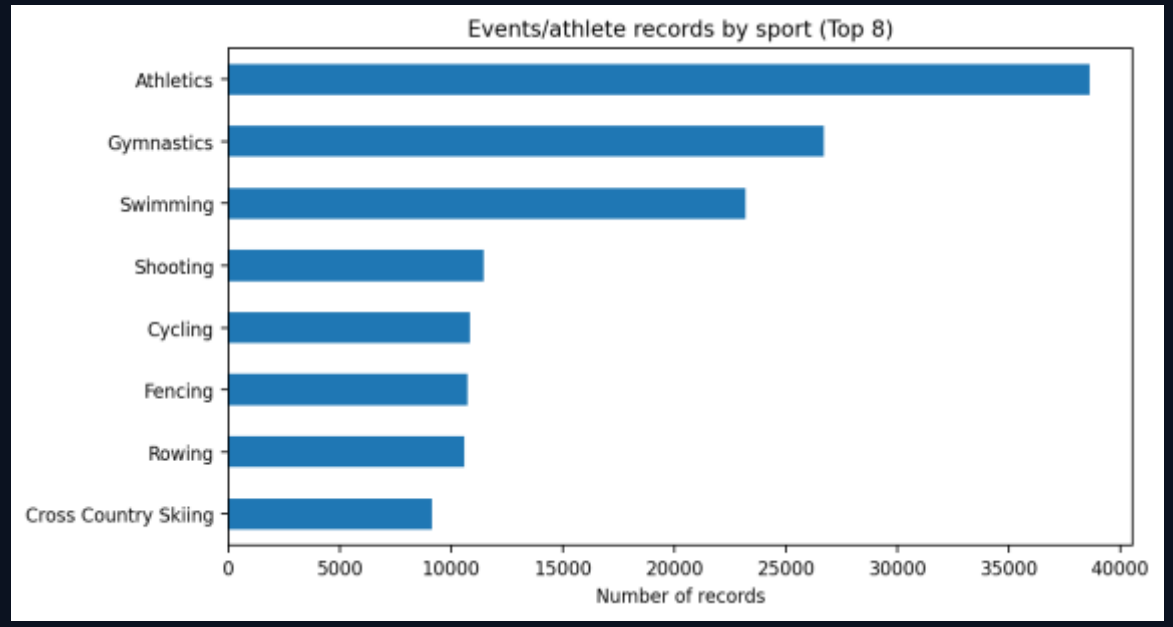
What the charts tell us at a glance

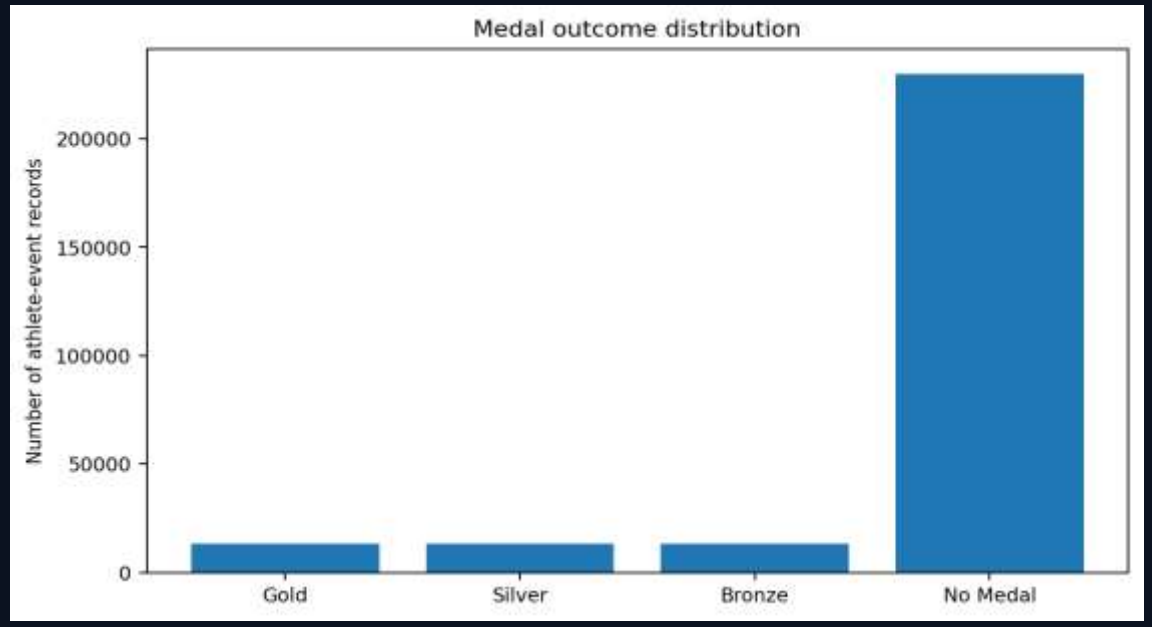
UNIQUE ATHLETES	FEMALE RECORDS	MEDAL RECORDS	SPORTS
134,732	27.6%	14.7%	66

- Athletics has the largest record count (38,624), followed by Gymnastics and Swimming.
- Most rows are “No Medal” (229,959); medal-winning rows total 39,772.
- The number of records varies across Olympic years; 1992 has the largest record count in this file.
- Female athlete-event records account for about 27.6% of all rows, indicating a gender representation gap across the full historical period.

06 | Deeper Insights

Patterns worth investigating further





- The United States has the highest count of medal-winning athlete-event records in this dataset (5,219), followed by the Soviet Union and Germany.
- Gold (13,369), Silver (13,108) and Bronze (13,295) counts are broadly similar.
- These are historical participation records. Host city, sport popularity, gender participation and medal outcomes should be explored by year/season to avoid misleading overall comparisons.

07 | Business Recommendations

How stakeholders can use the insights

01 Sponsorship strategy — Prioritize high-participation sports such as Athletics, Gymnastics and Swimming; validate audience reach before allocating budgets.

02 Athlete development — Compare medal outcomes by sport, country and Olympic year to identify areas for coaching investment.

03 Inclusion & access — Track female participation over time and by sport to identify gaps and measure progress.

04 Host-city planning — Use historical record volume by year/season to estimate staffing, venue and event-operation needs.

05 Better decisions — Add audience, viewership, costs, ticket sales and sponsorship revenue to connect sporting outcomes to actual business impact.

Business takeaway: use this dataset to spot patterns; combine it with financial and audience data before making investment decisions.